



LECTURE 5

DEEP REPRESENTATION LEARNING II

Learning beyond Memorization

PRESS **→** OR SPACE. 

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In this lecture

- Neural networks and learned representations
- Supervised training and optimization
- Learning useful intermediate features directly from data
- Convolutional neural networks
- Geospatial and remote-sensing examples

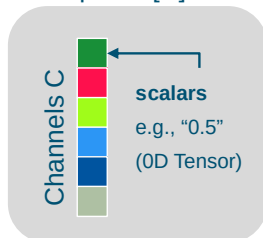
Data and Experience



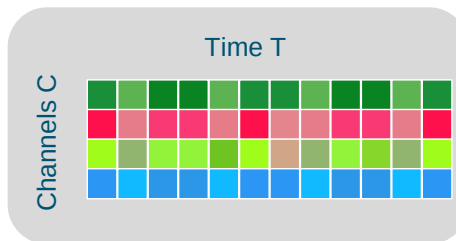
Data is the window through which a learner experiences the world.

Data as blocks of numbers - E.g., Tensors

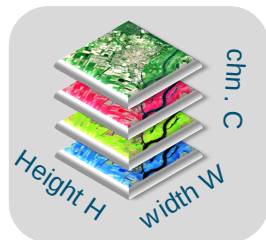
1D Tensor: Vector
e.g., pixels or
class probs [C]



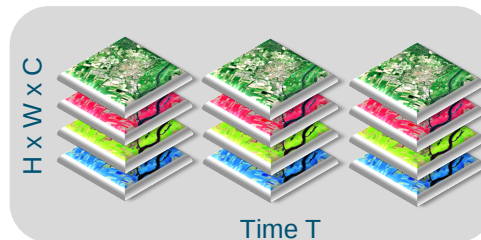
2D Tensor: Matrix
Time Series [T x C]



3D Tensors
Images [H x W x C]



4D Tensors
Img. Time Series [T x H x W x C]



Data samples from a distribution